

Teacher/Class: _____

DECEMBER 2006

Time Allowed: 70 minutes

- Write your name and class at the top of each page
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start **each** question on a **new** page.
- Diagrams unless otherwise stated are not to scale.

[illegible]

QUESTION 1 (7 Marks)**(MARKS)**

- a) Factorise $2x^2 + 5x - 3$ (1)
- b) Solve $x^2 - 4x > 0$ (2)
- c) Evaluate $\sum_{r=2}^5 \frac{1}{r+1}$ (1)
- d) Let α and β be the roots of $x^2 - 3x - 7 = 0$ find
- i) $\alpha + \beta$ (1)
- ii) $\alpha\beta$ (1)
- e) Form a quadratic equation whose roots are -2 and 3 (1)

QUESTION 2 (7 Marks) (start a new page)**(MARKS)**

- a) If $x = -2$ is a root of $4x^2 + x + k = 0$, find k (1)
- b) Find the limiting sum of the geometric series
- $$\frac{13}{5} + \frac{13}{25} + \frac{13}{125} + \dots$$
- (2)
- c) For the sequence $-8, -1, 6, \dots$
- i) Find the 29th term (1)
- ii) Find the sum of the first 29 terms (1)
- iii) Which term of the sequence has a value of 167? (2)

QUESTION 3 (7 Marks) (start a new page)**(MARKS)**

- a) i) Draw a neat sketch (using a ruler for the axes) of the parabola $x^2 = 8y$ (1)
- ii) Find the co-ordinates of the focus (1)
- iii) Find the equation of the directrix (1)
- iv) Find the equation of the tangent to the parabola $x^2 = 8y$ that passes through the point $(-8, 8)$ (2)

b) In solving a quadratic equation a student wrote his solution as

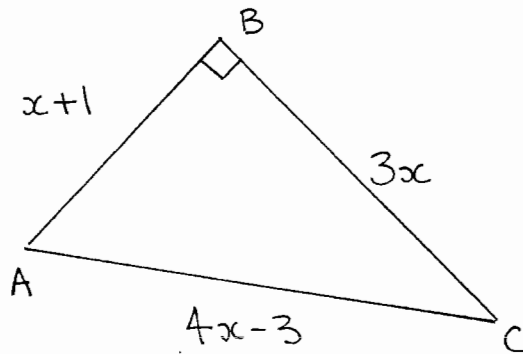
$$x = \frac{4 \pm \sqrt{16 + 96}}{6} \quad (2)$$

What was the original equation?

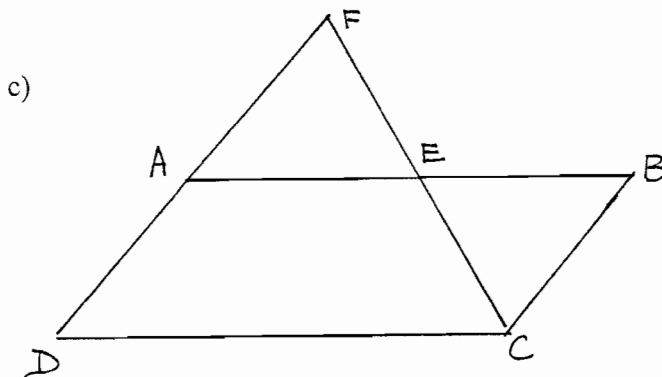
QUESTION 4 (7 Marks) (start a new page)

(MARKS)

- a) Triangle ABC below has angle ABC equal to 90° . Find all possible value(s) for x. (2)



- b) How much will \$500 grow to at 12% p.a. if compounded quarterly for 5 years (1)



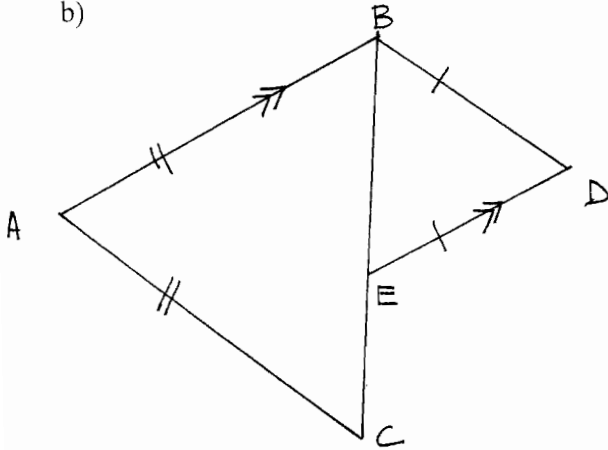
*ABCD is a parallelogram $AE=EB$
 DA produced intersects
 CE produced at F*

- i) Copy the diagram onto your answers sheet
- ii) Prove $\triangle AFE \cong \triangle BCE$ (2)
- iii) Hence explain why $DA = AF$ (2)

QUESTION 5 (7 Marks) (start a new page)**(MARKS)**

a) For what values of k does the equation $x^2 + kx + 3 - k = 0$ have real, different roots? (3)

b)



Triangles ABC and BDE are isosceles
 $AB = AC$ and $BD = ED$ and $AB \parallel ED$

- i) Copy the diagram onto your answer sheet.
- ii) Prove $\triangle ABC$ is similar to $\triangle BDE$ (3)
- iii) If $BD = 5\text{cm}$, $BE = 4\text{cm}$ and $AC = 6\text{cm}$ find the length of EC (1)

QUESTION 6 (7 Marks) (start a new page)**(MARKS)**

a) Insert three numbers between 5 and 80 so as to form five numbers in a geometric sequence . (3)

b) The first n terms of an arithmetic sequence have a sum given by $S_n = 25n - 2n^2$

- i) Find the first term and the second term (2)
- ii) Find the common difference (1)
- iii) Find the expression for the n th term (1)

QUESTION 7 (7 Marks) (start a new page)**(MARKS)**

a) A man places \$1500 at the beginning of each year into a superannuation fund, for 30 years. Interest on investments in the fund compounds at 12% p.a.

i) Find the amount he has in the fund at the end of 30 years (3)

ii) If this amount in part i) is taken as a lump sum and taxed at the rate of 30% for each dollar over \$50,000 , how much will he receive after tax. (2)

b) For the parabola $y = x^2 + 4x + 5$ find

i) the co-ordinates of the vertex (1)

ii) the co-ordinates of the focus (1)

QUESTION 8 (7 Marks) (start a new page)**(MARKS)**

a) A is the point $(8,0)$ and O is the origin. P is the variable point (x,y)

i) If P moves so that $PO = 3PA$, show that the locus of P is given by

$$x^2 + y^2 = 9[(x-8)^2 + y^2] \quad (2)$$

ii) Show that this locus is a circle by finding its centre and radius (2)

b) i) Find the sum of the geometric series

$$x^4 + x^3y + x^2y^2 + xy^3 + y^4 \quad (2)$$

ii) Hence factorise $x^5 - y^5$ (1)

QUESTION 1

a) $2x^2 + 5x - 3 = (2x - 1)(x + 3)$
 b) $x^2 - 4x > 0$
 $x(x - 4) > 0$



$\therefore x > 4 \text{ and } x < 0$

c) $\sum_{r=2}^5 \frac{1}{r+1} = \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}$
 $= \frac{19}{20}$

d) $a=1$ $b=-3$ $c=-7$

$\therefore$ i) $\alpha + \beta = 3$

ii) $\alpha\beta = -7$

e) $(x+2)(x-3) = 0$

QUESTION 2

a) sub $x=-2$ into $4x^2 + x + k = 0$
 $4(4) - 2 + k = 0$

$k = -14$

b) $a = \frac{13}{5}$ $r = \frac{1}{5}$

$\therefore S_{\infty} = \frac{\frac{13}{5}}{1 - \frac{1}{5}}$
 $= 3\frac{1}{4}$

c) $-8, 1, 6, \dots$ AP: $a = -8$
 $d = 7$

i) $T_{29} = -8 + 28 \times 7 = 188$

ii) $S_{29} = \frac{29}{2}(-8 + 188)$
 $= 2610$

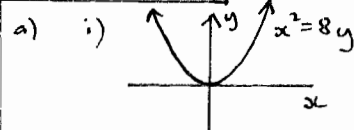
iii) $167 = -8 + (n-1) \times 7$

$167 = -8 + 7n - 7$

$182 = 7n$

$\therefore n = 26$ $\therefore T_{26} = 167$

QUESTION 3



ii) $4a = 8 \therefore a = 2$

Focus $(0, 2)$

iii) Directrix $y = -2$

iv) $x^2 = 8y$

$\therefore y = \frac{x^2}{8}$

$\frac{dy}{dx} = \frac{2x}{8} = \frac{x}{4}$

$\therefore m_T = -2$ at $(-8, 8)$

eqn. tang: $y - 8 = -2(x + 8)$

$y - 8 = -2x - 16$

$2x + y + 8 = 0$

b) $b = -4$ $a = 3$

$-4ac = 96$

$-12c = 96$

$c = -8$

$\therefore$ equation $3x^2 - 4x - 8 = 0$

(or any multiple of this)

QUESTION 4

a) $(4x-3)^2 = (x+1)^2 + (3x)^2$
 $16x^2 - 24x + 9 = x^2 + 2x + 1 + 9x^2$

$6x^2 - 26x + 8 = 0$

$3x^2 - 13x + 4 = 0$

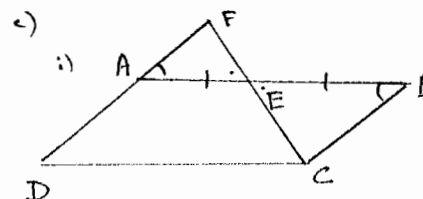
$(3x-1)(x-4) = 0$

$x = \frac{1}{3}$ $x = 4$

only valid answer $x = 4$

(since $4 \times \frac{1}{3} - 3 < 0$ hypot. -ve)

b) $A = 500 \left(1 + \frac{12.4}{100}\right)^{20}$
 $A = 500(1.03)^{20}$
 $A = \$ 903.06$



ii) In Δ 's AFE and BCE

$AE = EB$ (given)

$\hat{FAE} = \hat{EBC}$ (alternate angles)

$DF \parallel CB$, sides of parallelogram

$\hat{FEA} = \hat{BEC}$ (vertically opposite angles)

$\therefore \Delta AFE \equiv \Delta BCE$ (AAS)

iii) $AF = BC$ (corr. sides in congruent triangles)

$AD = BC$ (opp. sides of parm.)

$\therefore AD = AF$

QUESTION 5

a) $\Delta > 0$

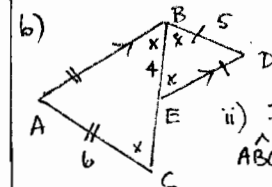
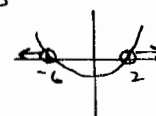
$k^2 - 4 \times 1 \times (3-k) > 0$

$k^2 - 12 + 4k > 0$

$k^2 + 4k - 12 > 0$

$(k+6)(k-2) > 0$

$k < -6$ and $k > 2$



ii) In ΔABC , B
 $\hat{ABC} = \hat{BED}$ (alt. angles)

[since $\hat{DBE} = \hat{DEB}$ opp. angles in iso. tria.]
 $\hat{ABE} = \hat{BCA}$

$\hat{DBE} = \hat{ACB}$ (base angles of iso. triangle)

$\therefore \Delta ABC \parallel \Delta BDE$ equian.

iii) $\frac{5}{6} = \frac{4}{BC}$ (corr. sides similar tri.)

$BC = 4.8$

$\therefore EC = 0.8$

QUESTION 6

a) $5 \dots 80$

$a = 5$ $T_5 = 80$

$80 = 5 \cdot r^4$

$r^4 = 16$

$r = \pm 2$

$\therefore$ AP $5, 10, 20, 40, 80$

OR $5, -10, 20, -40, 80$

b)

$S_n = 25n - 2n^2$

i) $S_1 = T_1 = 23$

$S_2 = 50 - 8 = 42 \therefore T_2$

$\therefore T_1 = 23$ $T_2 = 19$

ii) $d = -4$

iii) $T_n = 23 + (n-1) \times -4$

$T_n = 27 - 4n$

Question 7

a) i)

$$A = 1500(1.12)^{30} + 1500(1.12)^{29} + \dots + 1500(1.12)^1$$

$$= 1500(1.12^1 + 1.12^2 + \dots + 1.12^{30})$$

[C.P. $a = 1.12$ $r = 1.12$ $n = 30$]

$$A = 1500 \cdot \left[\frac{1.12(1.12^{30} - 1)}{1.12 - 1} \right]$$

$$A = \underline{\underline{\$405,438.91}}$$

ii) TAX = $.3 \times 355438.91$

$$= \$106631.67$$

$\therefore$ after TAX super = $\underline{\underline{\$298807.24}}$

b) i) $x^2 + 4x + 4 = y - 5 + 4$

$$(x+2)^2 = y - 1$$

vertex $(-2, 1)$

ii) $4a = 1 \therefore a = \frac{1}{4}$

focus $(-2, 1\frac{1}{4})$

Question 8

a) i) $PO = 3PA$

$$\sqrt{(x-0)^2 + (y-0)^2} = 3\sqrt{(x-8)^2 + (y-0)^2}$$

$$x^2 + y^2 = 9[(x-8)^2 + y^2]$$

ii) $x^2 + y^2 = 9(x^2 - 16x + 64 + y^2)$

$$x^2 + y^2 = 9x^2 - 144x + 576 + 9y^2$$

$$-576 = 8x^2 - 144x + 8y^2$$

$$-72 = x^2 - 18x + y^2$$

$$-72 + 81 = (x^2 - 18x + 81) + y^2$$

$$9 = (x-9)^2 + y^2$$

b) $x^4 + x^3y + x^2y^2 + xy^3 + y^4$

i) $a = x^4$ $r = \frac{y}{x}$ $n = 5$

$$S_5 = \frac{x^4 \left[\left(\frac{y}{x} \right)^5 - 1 \right]}{\frac{y}{x} - 1}$$

$$S_5 = x^4 \left[\frac{y^5 - x^5}{x^5} \right] \div \left(\frac{y-x}{x} \right)$$

$$= \frac{x^4 (y^5 - x^5)}{x^5} \times \frac{x}{(y-x)}$$

$$= \frac{-(x^5 - y^5)}{-(x-y)}$$

$$S_5 = \underline{\underline{\frac{x^5 - y^5}{x - y}}}$$

ii) $x^4 + x^3y + x^2y^2 + xy^3 + y^4$

$$= \frac{x^5 - y^5}{x - y}$$

$$\therefore x^5 - y^5 = (x-y) \left[x^4 + x^3y + x^2y^2 + xy^3 + y^4 \right]$$